$2 and tails costs you $1. How much of your bankroll should you bet if you're offered these odds?

According to the Kelly Formula, the edge is $0.50 [(0.5 × $2) + (0.5 × −$1)]. The odds are what you win, if you win, or $2. So the Kelly Formula suggests you bet 25 percent ($0.50/$2.00) each time. The first example involves more than 2 outcomes. For a detailed treatise on how to calculate the Kelly bet size for such bets, go to [www.cisiova.com/betsize.asp](http://www.cisiova.com/betsize.asp). This web site not only gives the general case Kelly Formula, but the author has generously programmed the formula for use by anyone at no charge. The intersted reader may also wish to read Edward Thorp's paper, "The Kelly Criterion in Blackjack, Sports Betting and the Stock Market." For the first example, the answer is 89.4 percent of your $10,000 bankroll or $8,940.

Papa Patel had likely never heard of the Kelly Formula. In Chapter 1, we noted that when Papa Patel invested $5,000 in his first motel, he pretty much bet it all on this investment. The odds in the aforementioned example are roughly the odds Papa Patel was offered—an 80 percent chance of having a 21 bagger, a 10 percent chance of a 7.5 bagger, and a 10 percent chance of going broke. In reality, Papa Patel was more conservative in his bet than the Kelly Formula suggested. He bet 50 percent of his bankroll. He did have $5,000 to his name and "bet it all." But, he had that ace in the hole—the ability to go back, take a job, save $5,000, and try again in a few years. He likely would not do this endlessly because each time he gets older and gets dissuaded from the endless bitter experiences. Because Dhandho is so deeply rooted in his psyche, he's got at least two bets in him. He puts 50 percent of his bankroll at risk on the first bet. If it works, he does not place a second bet. If it fails, he places a second bet.

Winning the first bet changes the world around him. His family no longer lives in the motel. They have hired help and can buy a bigger motel. When he now buys another motel (and hence places his second bet), it's with a smaller percentage of his bankroll because the odds are no longer as good. Even if the odds were simply a 50 percent probability of a 200 percent return and a 50 percent probability of a total loss, the Kelly Formula suggests that he ought to bet 25 percent of his bankroll.